

Data Centers: The host's terms

A data center brings a town new demand for power, housing, workers and water, and a large new tax base. Whether residents come out ahead depends on whether supply keeps up with that demand and on who collects the tax, and local and state rules decide both. In Quincy, Washington, the property tax rate on a home fell from \$14.81 per \$1,000 in 2005 to \$8.17 in 2026.

The usual objections to data centers, taken one at a time as case studies and checked against auditors' reports, levy files, county budgets and grid reports.

The objections to data centers, and the rule behind each outcome

THE THROUGH-LINE

Nearly every complaint is about the new demand a campus brings for electricity, housing, labor and water, while most of a town's gain comes through the property tax. Each section can be read with two questions in mind, whether supply kept up and who collected the tax. Homeowners in Quincy and Loudoun came out ahead; renters in Abilene and ratepayers in Virginia paid more.

WHY NOW

Gallup found 71 percent of Americans would oppose an AI data center in their area, and by August towns in 42 states had adopted or proposed more than 500 local moratoria. The costs we found fall mostly on renters and a campus's nearest neighbors. A moratorium avoids them, and it also gives up the tax and the chance to set the terms.

WHAT YOU'LL LEARN

1. The claim of little local return: what Quincy's growing tax base did to its rate (§01)
2. The poverty series: what the census numbers can and can't tell us (§02)
3. The tax contribution and the tax break: the same equipment taxed at \$0.66 to \$4.57 across Virginia's counties (§03)
4. Few permanent jobs and temporary work: a crew that arrived before the housing, in Abilene (§04)
5. The Cerrillos figure, and the question of water: what a town can learn before it approves a campus (§05, §06)
6. The price of power and emissions: how fast Texas and PJM connect new plants (§07, §08)
7. Noise, generators and opposition: what localities could regulate, and what a moratorium gives up (§09, §10)

IF YOU REMEMBER ONE LINE: Towns came out ahead where supply kept up and the town kept the tax. Watch PJM's 2029/30 auction, Abilene's rents in 2027 and Loudoun's fiscal 2028 budget.

Local rules and local outcomes

Residents came out ahead where the rules let power and housing keep up with a campus's demand and let the town keep the property tax. Where supply came late, as with Abilene's housing and Virginia's power plants, renters and ratepayers paid more.

Second Order · Issue 12 claim

\$14.81 to \$8.17

combined property tax rate per \$1,000 on a Quincy home, 2005 and 2026, school and hospital levies included. On a \$300,000 home, that's \$4,443 a year against \$2,451.

40.3%

of Loudoun County's fiscal 2027 general fund comes from data center buildings and equipment, and the real estate rate fell from \$1.205 to \$0.805 per \$100.

24 vs 41 months

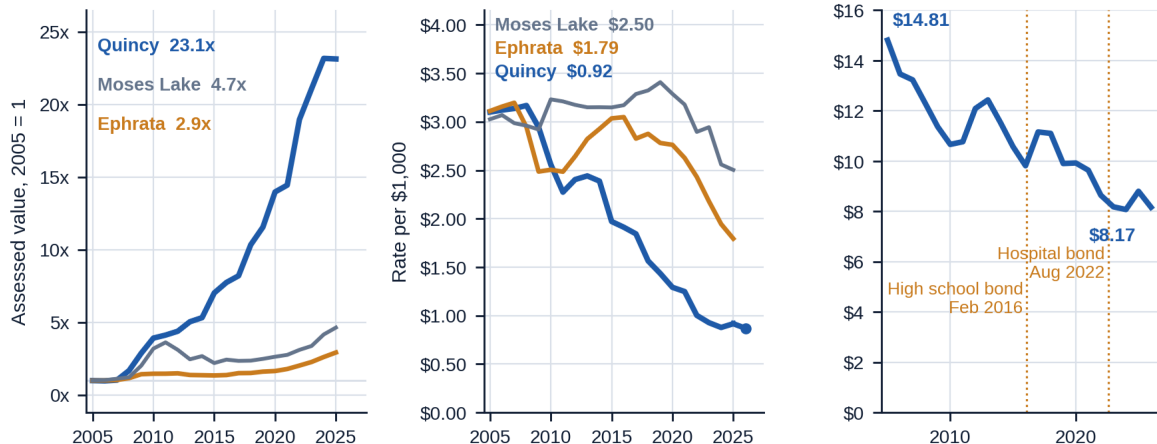
average wait from connection request to agreement in ERCOT, against network service requests elsewhere, on Lawrence Berkeley's 2026 survey. Texas added 14 gigawatts in 2024.

01 · EVIDENCE

EXHIBIT 1

Quincy's taxable value outgrew its levy, and its tax rate fell

Assessed value indexed to 2005; city levy rate and the combined rate a Quincy homeowner pays, dollars per \$1,000 of value. Taxes due 2005 to 2026



Note: The growth panel indexes each city to its own 2005 assessed value, so the three growth multiples share one axis; the dollar endpoints are labelled and nominal. Washington caps each district's regular levy at 1 percent growth a year plus new construction, so a district's rate falls when assessed value grows faster than the allowed levy. The city comparisons are descriptive only, and Moses Lake also has data centers. The combined rate is for tax code area 23, and the earlier totals assume the same districts covered the area. Household bills also depend on assessments.

Source: Washington Department of Revenue, Local Taxing District Levy Detail, 2005 to 2025; Grant County Assessor levy sheets for 2026; Second Order calculations.

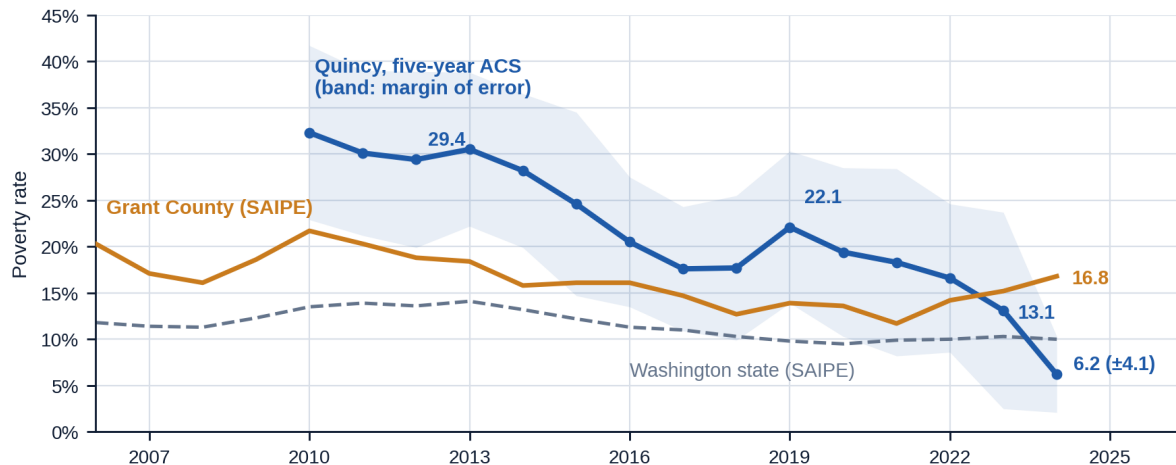
SO WHAT

- Quincy's city levy rate fell 72 percent from 2005 to 2026 as its tax base grew 23 times by 2025, about fourteen times after inflation. Ephrata, the county seat, under the same 1 percent cap, cut its rate 42 percent on a base that grew 2.9 times.
- The combined rate fell from \$14.81 to \$8.17 in tax code area 23, even though voters passed a \$108.3 million school bond in 2016 and a \$55 million hospital bond in 2022.
- The hospital bond costs about 35 cents per \$1,000 on the 2026 levy sheet, less than the 46 to 48 cents projected when it passed, because the tax base kept growing.

EXHIBIT 2

Quincy's estimated poverty rate fell further than Grant County's

Share of residents below the poverty line, percent. Quincy: ACS five-year estimate, plotted at the last year of each window, with its 90 percent margin of error. County and state: Census SAIPE, annual



Note: Adjacent five-year windows are not independent. Comparing the separate 2008 to 2012 and 2020 to 2024 windows gives an estimated decline of 23.2 points, with a 90 percent margin of about 10.3 points that combines the uncertainty at both ends. The margin covers sampling uncertainty only, and the calculation does not identify whether data centers caused the decline. Quincy's first campus opened in 2007.

Source: Census Bureau ACS five-year summary files (B17001), vintages 2010 to 2024; Census SAIPE via FRED; Second Order calculations.

SO WHAT

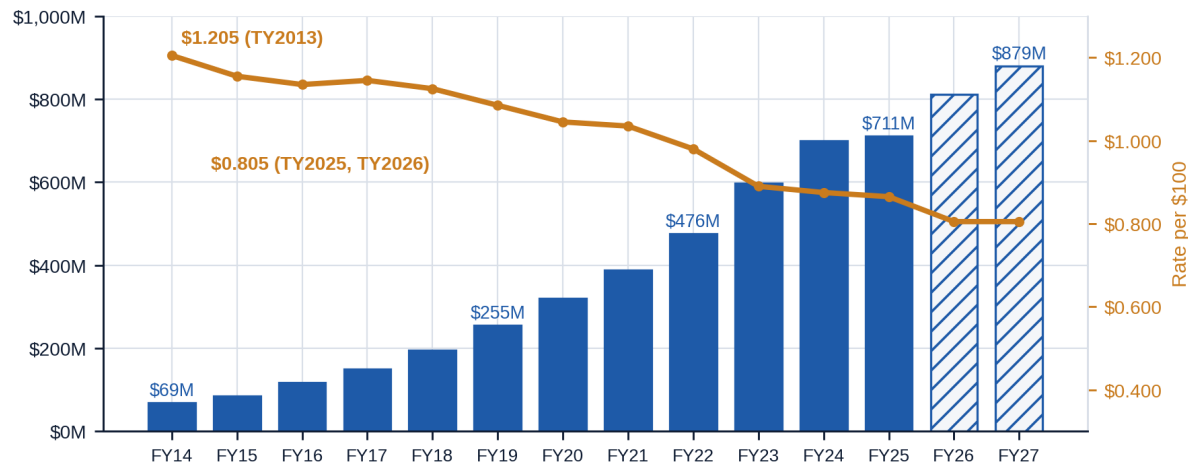
- Quincy's five-year poverty estimate went from 29.4 percent (2008 to 2012) to 22.1 (2015 to 2019) to 6.2 (2020 to 2024). That last figure has a 4.1-point margin of error, and the 2019 to 2023 estimate was 13.1.
- Grant County's model-based rate was 21.7 in 2010 and 16.8 in 2024, higher than any year since 2017. It's a different measure from Quincy's five-year series.
- The estimates point to a drop of about four fifths, and no one has measured how much of it the campuses caused.

03 · EVIDENCE

EXHIBIT 3

Loudoun cut its real estate rate by a third while its equipment tax receipts rose more than twelvefold

Bars: computer equipment personal property tax revenue, \$ millions, by fiscal year (actuals to FY2025; FY2026 and FY2027 adopted, the latter data center filers only). Line: real property tax rate, dollars per \$100, by tax year



Note: The fiscal 2014 starting figure covers all computer equipment, while the fiscal 2027 figure counts only data center filers. The broader starting figure makes the implied growth in data center receipts conservative, and no exact like-for-like multiple is available. From fiscal 2022 data centers account for 97 percent or more of the line. Fiscal 2026 and 2027 are adopted budget figures, the latter for data center filers (\$879 million). Data centers supplied 38 percent of the fiscal 2026 general fund.

Source: Loudoun County adopted budgets, fiscal 2016 to 2027; county data center page; Second Order compilation.

SO WHAT

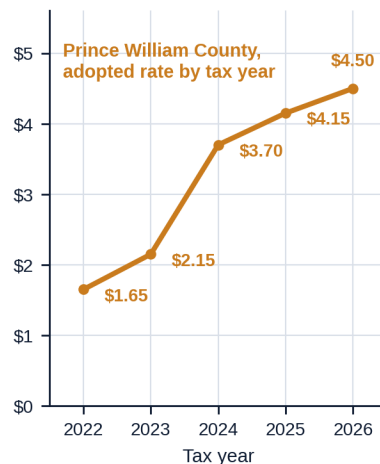
- Loudoun's computer equipment tax brought in \$69 million in fiscal 2014, before data centers were counted separately. Data centers paid \$700 million in fiscal 2025, and the county expects \$879 million in fiscal 2027.
- The real estate rate fell from \$1.205 per \$100 in tax year 2013 to \$0.805 in 2025 and stayed there for 2026, the first year without a cut in a decade.
- Data center buildings and equipment make up 40.3 percent of the fiscal 2027 general fund. The state's sales tax break on the same equipment cost \$1.6 billion in fiscal 2025, in the state's accounts, not the county's.

03 · EVIDENCE

EXHIBIT 4

Under one Virginia law, counties tax the same equipment at \$0.66 to \$4.57

Left: tax rate on data center computer equipment, dollars per \$100 of assessed value, by county, latest adopted year; the marker is the county's general personal property rate, the statutory ceiling. Right: Prince William's adopted rate by tax year



Note: Rates are per \$100 of assessed value on computer equipment and peripherals used in a data center, a class for which Virginia lets each locality set a rate up to its general personal property rate. Loudoun, Prince William, Fairfax and Culpeper charge their maximum; Henrico and Mecklenburg charge less. Prince William's series comes from its adopted budgets for tax years 2022 to 2026; an earlier \$1.25 rate, reported only in the press, is left out. Henrico's \$0.40 rate ran from tax year 2017 to the fiscal 2025 schedule.

Source: Code of Virginia 58.1-3506; Loudoun FY2027 adopted budget; Prince William County budget adoption releases, 2022 to 2026; Henrico approved budgets FY2019, FY2022 and FY2026; Fairfax Department of Tax Administration; Culpeper County; Mecklenburg County FY2025 tax ordinance; Second Order compilation.

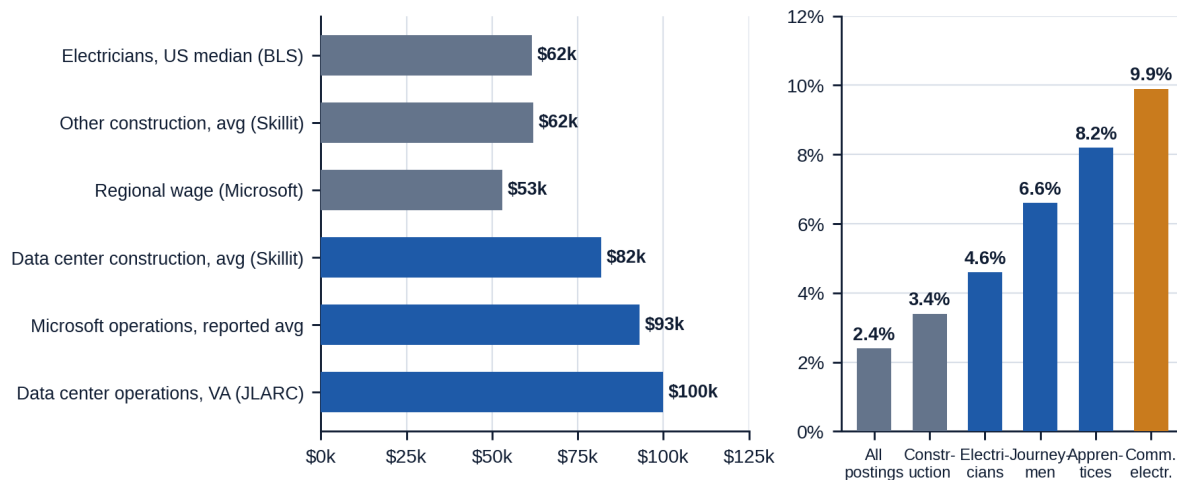
SO WHAT

- Virginia treats data center computer equipment as its own tax class and lets each county set the rate, up to its general rate. Loudoun charges \$4.15 per \$100, Fairfax \$4.57 and Culpeper \$3.50.
- Prince William raised its rate from \$1.65 in tax year 2022 to \$4.50 for 2026, four budgets in a row. Henrico cut to \$0.40 in 2017 to attract a campus and went back up to \$2.60 for 2025. Mecklenburg charges \$0.66.
- A county can't go above its general personal property rate, which is why Prince William stopped at \$4.50. Even at the low rate, Henrico's data center revenue seeded a \$60 million housing trust fund in 2024.

EXHIBIT 5

The few permanent jobs pay at or above the local average, and data center contractors are bidding up electricians' pay

Left: reported annual pay, dollars, for data center operations and construction work against the baselines each source used. Right: year-on-year advertised-pay growth, May 2025 to May 2026, percent, by category



Note: Left: reported averages and one median, from different sources and years; single-site reports of much higher pay are discussed in the text and left off the chart. Microsoft's figures are company reports. Right: growth in advertised pay on Indeed, via Bloomberg; advertised pay does not track actual raises for the same workers.

Source: JLARC Report 598 (December 2024); Hamm Institute (November 2025); Skillit via Fortune (December 2025); Microsoft (May 2026); Census ACS 2020 to 2024; Texas Tribune (April 2026); Bloomberg via Insurance Journal (July 2026); BLS OEWS; Second Order compilation.

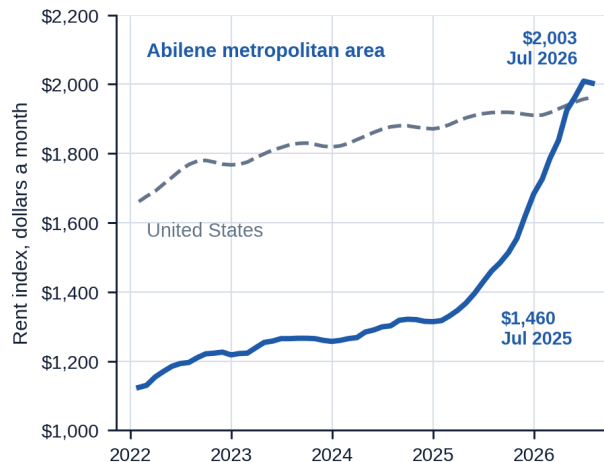
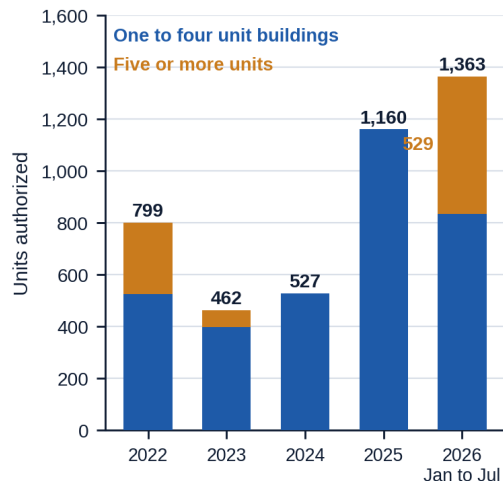
SO WHAT

- JLARC counts about fifty workers per building earning around \$100,000, half of them contractors such as electricians, pipefitters and security. Microsoft pays its data center staff \$93,000 against a regional \$53,000. No source records where the workers live.
- Construction makes up 80 percent of the modeled Virginia jobs and labor income. Skillit puts data center construction pay at \$81,800 against \$62,000 on other projects, a 32 percent premium.
- Advertised electrician pay on Indeed rose 4.6 to 9.9 percent in the year to May 2026, against 2.4 percent for all jobs, and one Abilene builder says his houses now take two months longer.

EXHIBIT 6

Abilene's permits doubled as rents rose 37 percent, and the apartments permitted in 2026 are not yet built

Left: housing units authorized by permit, Abilene metropolitan area. Right: Zillow's rent index, dollars a month, Abilene and the United States



Note: Left: housing units authorized by building permits in the Abilene metropolitan area (Callahan, Jones and Taylor counties), yearly for 2022 to 2025 and January to July for 2026, from the Census Bureau's Building Permits Survey. The 2026 bar includes 529 units in buildings of five or more, after none in 2024 and 2025. Right: Zillow's smoothed monthly rent index for the Abilene area and for the United States, a repeat-rent index of asking rents; KUT reported an average of listings. Permits are authorizations to build and come before completions.

Source: Census Bureau Building Permits Survey; Zillow Observed Rent Index (ZORI); Second Order calculations.

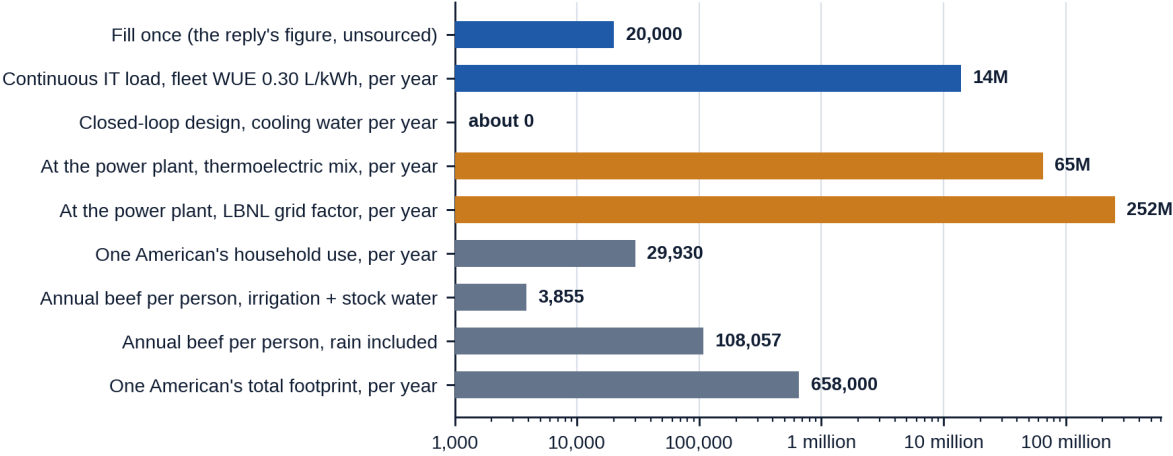
SO WHAT

- Abilene cut the campus's property tax by 85 percent for ten years, the longest Texas allows. The deal sets minimums for jobs and wages and contains nothing about housing.
- Zillow's rent index for the area went from \$1,460 in July 2025 to \$2,003 in July 2026, up 37 percent, and passed the national index in May 2026. KUT's \$2,600 figure is an average of listings.
- Builders got permits for 527 units in 2024, 1,160 in 2025 and 1,363 in the first seven months of 2026, including 529 apartments after none in two years. Permits rose with the rents, but the apartments won't open until 2027.

EXHIBIT 7

Cooling design sets a campus's water demand, and its electricity adds indirect water use at power plants

Gallons attributable to a 20 megawatt (IT) cluster on each basis, log scale, against one American's beef and household water. Second Order calculations



Note: The circulating fill volume is unverified and is not used to calculate water savings; zero refers to cooling only. The chart assumes continuous IT use and a PUE of 1.2; a PUE of 1.1 or 1.3 moves the power-plant estimates by about 8 percent. The fleet WUE and the LBNL grid factor are averages across designs or power mixes. The LBNL factor includes reservoir evaporation and is not an upper bound.

Source: Microsoft (December 2024); USGS SIR 2019-5103; LBNL 2024 energy usage report; Mekonnen and Hoekstra (2012); USDA ERS; USGS Circular 1441; Second Order calculations.

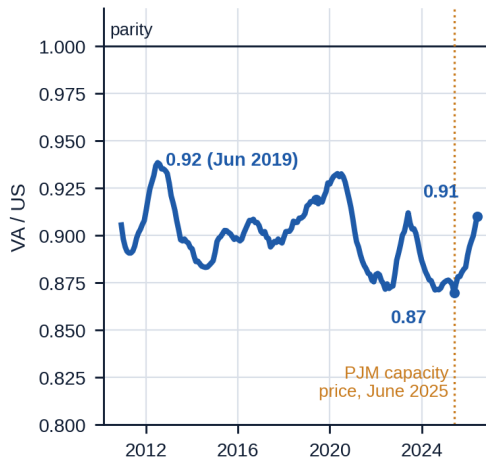
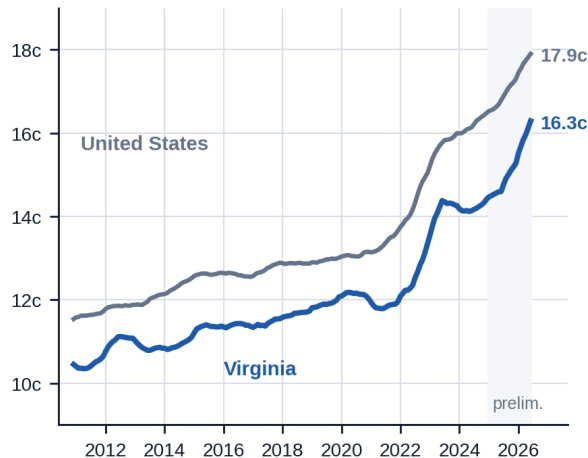
SO WHAT

- The often-repeated claim that a 20 megawatt cluster is filled once with 20,000 gallons has no public source.
- At Microsoft's fleet average, a 20 megawatt IT load running around the clock uses about 14 million gallons a year. Microsoft, Oracle and Vantage have published closed-loop designs meant to use no water for cooling.
- The power plants serving the same load use about 65 million or 250 million gallons a year, depending on the power mix assumed, five to eighteen times what the campus uses for cooling.

EXHIBIT 8

Virginia's household electricity price has stayed below the national average since 2010

Left: residential electricity price, cents per kilowatt-hour, trailing twelve months, sales-weighted. Right: Virginia as a share of the national figure. EIA-861M, December 2010 to June 2026



Note: The first capacity price above \$100 per megawatt-day (\$269.92 for 2025/26, against \$28.92 the year before) took effect June 1, 2025. The causal estimate of about minus 6 percent (Watten, Bistline and Blanford, June 2026) covers 2019 to 2024 and ends before that date. Data from January 2025 on are preliminary.

Source: EIA-861M (August 26, 2026 release); PJM base residual auction results; Second Order calculations.

SO WHAT

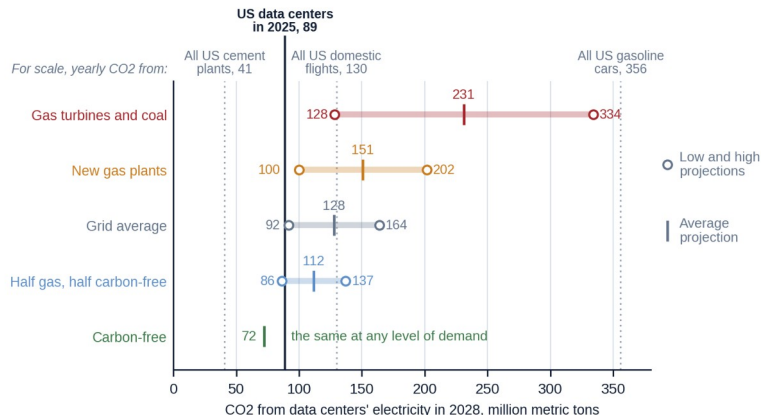
- Watten, Bistline and Blanford estimate that rates were about 6 percent lower from 2019 to 2024 than they would have been without data center growth, and the result holds up under stricter statistical tests. Their data ends before PJM's first high capacity price.
- Virginia's residential price has been below the national figure every month since January 2010. The ratio was 0.92 in June 2019, 0.87 in June 2025 and 0.91 in June 2026.
- No regulator has broken Virginia's 11.6 percent rise down by cause. PJM says 74 percent of the projects it has studied since 2020 withdrew, and only 23 of the 103 gigawatts with signed agreements are running.

EXHIBIT 9

Whether data center emissions rise or fall by 2028 depends on the power built to serve new demand

The five cases differ only in the power that serves growth after 2025

Only carbon-free supply keeps 2028 emissions under the 2025 figure across the whole demand range



Note: Second Order calculation. Use through 2023 is Lawrence Berkeley's estimate, with 2015 to 2017 and 2019 to 2022 read from its chart, and 2024 and 2025 extend it at the IEA's growth rates. Emissions are use times the power sector's CO₂ per megawatt-hour, which puts 2023 at 62 million tons against Lawrence Berkeley's 61 and follows EIA's 2026 outlook after 2025. Every 100 million tons is about 2 percent of US energy-related CO₂ in 2025. In every case load up to the 2025 level gets the grid average, and only growth beyond it gets the case's rate. The rates, in metric tons per megawatt-hour, are 0 for carbon-free power, 0.20 for half gas and half carbon-free, 0.40 for new combined-cycle gas and 0.81 for an even split of gas turbines and coal. The turbine rate uses EIA's tested heat rate, which understates emissions in real operation. The IEA's 2030 figure starts from a lower 2023 level, 154 terawatt-hours. The comparison lines are cement production (2023) and domestic commercial aircraft and gasoline passenger cars (2022) from EPA's greenhouse gas inventory.

Source: Lawrence Berkeley National Laboratory, 2024 United States Data Center Energy Usage Report; IEA, Key Questions on Energy and AI (2026); EIA, Monthly Energy Review tables 7.2b, 11.1 and 11.6, Annual Energy Outlook 2026, Electric Power Annual tables 8.1 and 8.2, CO₂ emission coefficients. EPRI, Powering Intelligence (2026); McKinsey (2024); EPA, Inventory of US Greenhouse Gas Emissions and Sinks, tables 2-1 and 3-13

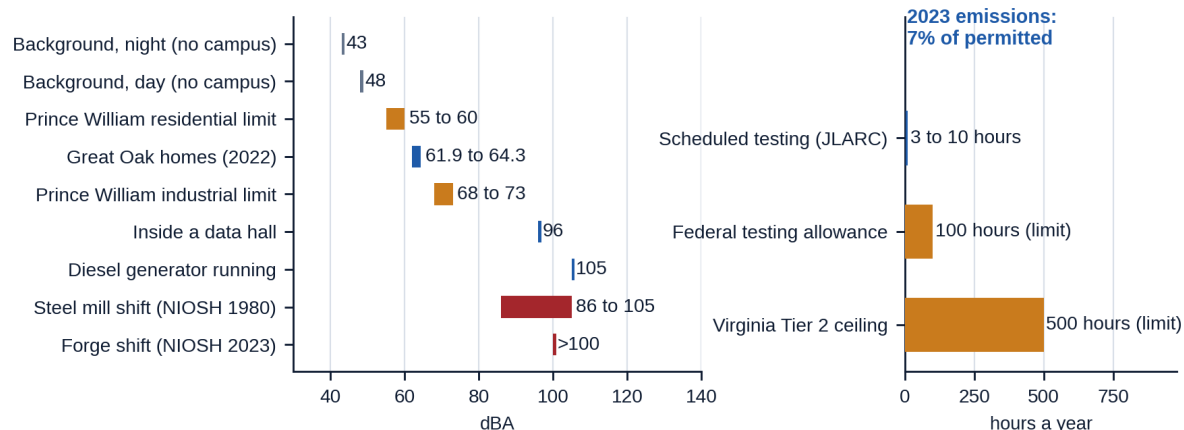
SO WHAT

- US data centers used 4.4 percent of the country's electricity in 2023 and emitted about 61 million tons of CO₂, a little over 1 percent of US energy-related emissions.
- If the growth after 2025 runs on carbon-free power, 2028 emissions come to about 72 million tons. With gas turbines and coal, they reach about 230 million, more than every US domestic flight.
- About 3 percent of the supply Texas added in 2024 and 2025 was gas, and Dominion plans 8.5 gigawatts of new gas. Virginia's and Ohio's large-load tariffs make data centers pay for most of what's built for them for 12 to 14 years.

EXHIBIT 10

Virginia localities lacked authority to set a low-frequency noise limit, and scheduled generator testing takes few hours

Left: sound levels in A-weighted decibels. Right: generator testing hours a year (blue) and permit limits (amber)



Note: Workplace and residential measurements are different exposures. Great Oak is a citizen study, and its readings come from different sites than the background readings. Testing hours describe a schedule reported in interviews. Permit limits cover different uses, and emissions figures do not measure hours of operation.

Source: JMT for Prince William County (2025); Prince William ordinances (2023, 2025); Lyver (2022); EESI; NIOSH (1980, 2023); Virginia JLARC (December 2024); Virginia DEQ via PEC; EPA; Environment America; New York Times (2012).

SO WHAT

- One resident's driveway readings beside the Manassas campus were 61.9 to 64.3 dBA. The background readings of 43 to 48 came from other places, so the gap isn't a measured campus effect. The ordinance uses a scale that discounts the 90 hertz tone.
- The county's consultant confirmed a 90 hertz tone likely audible indoors. JLARC found localities had no clear authority to set a low-frequency limit. Chandler caps noise at the level measured before construction, and Loudoun ended by-right approval in March 2025.
- JLARC describes maintenance and annual testing that add up to about 3 to 10 scheduled hours a year, and actual 2023 generator emissions were 7 percent of what the permits allowed.

Five findings

1. The tax rate

Quincy's rate fell because Washington's levy cap forces rates down as taxable value grows. Loudoun taxes equipment at its general rate of \$4.15 and expects \$879 million in fiscal 2027. Henrico charged \$0.40 for eight years, and Mecklenburg charges \$0.66 under the same law.

2. The abatement

Abilene cut the campus's tax by 85 percent for ten years with nothing about housing. Permits doubled in 2025 and kept rising in 2026 while rents rose 37 percent. On Oracle's estimate, the city keeps about \$30 million a year and gives up about \$170 million.

3. The connection rule

ERCOT checks a plant for reliability only, charges it nothing for network upgrades and reached agreements in 24 months on average, against 41 for network service elsewhere. Three quarters of PJM's projects withdrew, and its capacity auctions have hit the cap since the 2026/27 auction.

4. Water and noise

Karen Hao's Cerrillos figure was a thousand times too high because of a units mix-up in the filing. A campus that meets an ordinance written in A-weighted decibels can still be heard indoors at 90 hertz. Generator emissions ran at 7 percent of what permits allowed.

5. The permit

The Dalles took \$28.5 million of Google-funded water infrastructure, then sued The Oregonian to keep Google's water use secret. Minnesota's permitting agency now has to consider closed-loop cooling. A moratorium gives up the property tax, \$1.3 billion in Loudoun's fiscal 2027 budget.

Valuing a network from one county

Newton County, Indiana, could assess only its two miles of track, while much of the railroads' value showed up in the price of farms across the country. Grange farmers, whose land the network had made more valuable, fought the railroads over freight rates, and regulation followed while the railroads kept laying track. A county values a data center by the building, the servers and the jobs inside the county, while the value of the computing shows up in what buyers elsewhere produce. We expect the argument to move from local objection to state and federal terms, and later to limits on taxing campuses that can't move.

About a third

what American farmland in 1890 would have been worth without the railroads, on Donaldson and Hornbeck's estimate

1887

the Interstate Commerce Act requires railroads to publish rates that are "reasonable and just"

1976

the 4-R Act bars states from taxing railroads more heavily than other businesses

WATCHLIST

What would change our view

The Virginia ratio: Exhibit 8's Virginia-to-national price ratio, 0.91 in June 2026 and below 1.0 every month since January 2010, in EIA's monthly filings through 2027. The full 2027 year comes out in the spring of 2028.

Loudoun's fiscal 2028 proposal: early 2027: how much it leans on data center revenue, what it holds in reserve and what it projects for household bills.

The next ACS release: The ACS 2021 to 2025 release in December 2026, with its margin of error. It shares four years with the last estimate, and the levy finding doesn't depend on the poverty numbers.

PJM's next auction and Abilene's rents: PJM's next capacity auction, for 2029/30, the first sign of whether the queue cycle opened in April 2026 has added capacity. A fourth result at the cap would mean it hasn't yet. And Abilene's rent index through 2027, as the 529 apartments permitted in 2026 open.

Mansfield and Manassas: What the Mansfield, Georgia court hears and any ruling on the merits in the four families' suit against Meta, plus Prince William's first year of measurements under its October 2025 ordinance.